Principle of Temporal Lensing (PTL): A Neurocomputational Framework for Quantifying Subjective Time Distortion During Autobiographical Recall

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Abstract

Subjective time distortion during autobiographical recall is a robust psychological phenomenon without a unified quantitative account. The Principle of Temporal Lensing (PTL) provides a biologically anchored, mathematically closed **framework**—not a fully derived theory—in which perceived recall duration emerges as an informational functional defined over trajectories in neural–cognitive state space. The foundational metric choice, the Fisher information metric, is uniquely justified under reparametrization invariance by Chentsov's theorem, applied to the statistical manifold of neural recall states as defined in Definition 1. This is a modeling assumption with strong theoretical justification, not a derivation from first principles independent of that assumption.

Four empirically lockable channels govern the distortion: fractional memory persistence (M^*), affective charge geometry (E^*), recursive neural entropy (R), and narrative coherence (C). All parameters are fixed at the cohort level from preregistered empirical anchors; no individual-level fitting is permitted. The affective channel contributes a bounded multiplicative factor via $(1 + E^*)$, where the factor of 2 at maximum affect is a direct consequence of SAM instrument bounds, not an independently derived quantity. The framework generates five falsifiable predictions: (P1) a biological saturation ceiling on time dilation; (P2) power-law trauma memory persistence modeled by Mittag-Leffler dynamics; (P3) narrative reorganization causally alters retrospective duration; (P4) dissociative recursion entropy compresses subjective time; and (P5) narrative coherence independently modulates perceived duration under matched arousal. PTL is described as a framework throughout; the

designation of "theory" is earned only upon empirical validation of these predictions. The framework integrates information geometry, fractional calculus, and affective psychophysics into a single estimator-consistent equation with no individual-level free parameters.

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1 Introduction

Clock time and psychological time diverge systematically. A traumatic event recalled for sixty seconds of objective duration may be experienced as far longer; a flow state of equal clock duration may feel compressed to moments. This divergence is not a measurement artifact—it is a replicable feature of human memory and affect with well-documented neural correlates (Zakay & Block, 1992; Tse et al., 2004; Droit-Volet & Meck, 2007).

Existing accounts fall into two classes. Prospective models track time as it passes, emphasizing attentional resources and pacemaker-accumulator mechanisms (Zakay & Block, 1992; Gibbon, 1977; Meck, 1983). Retrospective models account for duration judgments after the fact, emphasizing memory density and contextual change (Ornstein, 1969; Block, 1990; Block & Gruber, 2014). Neither class produces a closed quantitative equation linking measurable neural and psychological variables to a specific duration estimate. Parameters are tuned post hoc or left implicit.

PTL addresses this gap. The core claim is that perceived recall duration is proportional to the informational path length traversed by the neural–cognitive system during a recall episode, governed by four observable channels anchored to preregistered empirical quantities. This claim rests on a foundational modeling assumption—that neural recall states form a statistical manifold to which the Fisher information metric applies—which is uniquely justified by Chentsov's theorem under the constraint of reparametrization invariance, but is not independently derived from first principles. PTL is a principled quantitative framework constructed under justified assumptions. It earns the designation of theory only upon empirical validation of the five predictions in Section 11.

The equation has no individual-level free parameters. Parameters are estimated at the cohort level before individual prediction. This distinguishes PTL from post-hoc curve fitting: cohort-level parameter locking means the model can fail for a given individual without adjustment, which is what makes individual predictions falsifiable.

The paper proceeds as follows. Section 2 establishes the statistical manifold formulation and its scope. Sections 3–4 derive the informational line element and Lagrangian action. Sections 5–8 derive each channel from first principles. Section 9 assembles the canonical equation. Section 10 proves estimator consistency. Section 11 formalizes the five predictions. Section 12 situates PTL within existing literature.

2 Neural Recall as a Statistical Manifold Process

Definition 2.1 (Neural Recall Distribution)**.** *Let autobiographical recall evolve as a stochastic process with distribution $p(x \mid \theta)$, parametrized by latent cognitive state vector*

$\theta \in \Theta \subseteq \mathbb{R}^d$.

Scope of Definition 1. This definition places neural recall states on a statistical manifold of probability distributions. This is a modeling assumption: it treats the neural system as a parametric family of distributions over observable neural activity x. The assumption is consistent with the information-geometric tradition in computational neuroscience (Amari, 1998; Pouget et al., 2000; Séries et al., 2004) and is adopted here as the foundational construction of the framework. All subsequent claims, including the Fisher metric justification, are conditional on this assumption.

Definition 2.2 (Statistical Manifold)**.** *The statistical manifold is*

$$\mathcal{M} = \{p(x \mid \theta) \mid \theta \in \Theta\},$$

equipped with the Fisher information metric

$$g_{ij}(\theta) = \mathbb{E}[\partial_i \log p(x \mid \theta)\, \partial_j \log p(x \mid \theta)]\,.$$

The Fisher metric g_{ij} is symmetric positive semi-definite by construction and positive definite under parameter identifiability. It measures the statistical distinguishability of nearby cognitive states: large g_{ij} indicates high informational sensitivity, meaning small perturbations in θ produce large distributional changes in neural dynamics.

Justification via Chentsov's Theorem. The Fisher metric is not chosen arbitrarily. Chentsov's theorem (Chentsov, 1982) establishes that the Fisher information metric is the *unique* Riemannian metric on a statistical manifold that is invariant under sufficient statistics (equivalently, under reparametrization of the sample space). Applied here: if perceived duration is to be independent of how we arbitrarily label neural states—a minimal well-formedness condition for any duration measure—then the Fisher metric is the unique valid choice given Definition 1. This is not a derivation from first principles independent of the modeling assumption; it is a derivation of uniqueness *within* that assumption. The scope is explicit and bounded.

Definition 2.3 (Recall Trajectory)**.** *A recall episode of objective duration t is a smooth curve*

$$\Gamma : [0, t] \to \mathcal{M}, \quad s \mapsto \theta(s), \quad \dot{\theta}^i = d\theta^i/ds.$$

3 Informational Line Element and Perceived Duration

The infinitesimal informational displacement along Γ is

$$d\ell^2 = g_{ij}(\theta)\, d\theta^i\, d\theta^j.$$

The total informational path length of a recall episode is

$$\ell = \int_0^t \sqrt{g_{ij}(\theta(s))\, \dot{\theta}^i(s)\, \dot{\theta}^j(s)}\, ds.$$

Definition 3.1 (Distortion Density)**.**

$$\Psi(s) = \sqrt{g_{ij}(\theta(s))\, \dot{\theta}^i(s)\, \dot{\theta}^j(s)}.$$

Proposition 3.2 (Proportionality Axiom)**.** *Perceived duration T' is proportional to informational path cost:*

$$T' = \alpha \int_0^t \Psi(s)\, ds = \alpha\, \ell.$$

The proportionality axiom is grounded in the contextual change hypothesis (Ornstein, 1969; Block, 1990): events feel longer when more distinct contextual states are traversed. The Fisher metric formalizes "contextual change" as statistical distinguishability between successive neural states. This axiom is a foundational modeling claim of the framework, not a consequence of prior empirical results alone.

4 Informational Lagrangian and Action

To decompose Ψ into measurable channels, we define an informational Lagrangian.

Definition 4.1 (Informational Lagrangian)**.**

$$\mathcal{L}(s) = I_M(s) + I_E(s) - I_R(s) - I_C(s),$$

where I_M and I_E are positive (dilation) contributions from memory density and affective charge, and I_R and I_C are negative (compression) contributions from recursive neural entropy and narrative coherence.

Definition 4.2 (Informational Action)**.**

$$S[\Gamma] = \int_0^t \mathcal{L}(s)\, ds.$$

Perceived duration is modeled as exponential trajectory gain over the time-averaged Lagrangian:

$$T' = \alpha \exp\left(\frac{1}{t}\int_0^t \mathcal{L}(s)\,ds\right).$$

The exponential form ensures $T' > 0$ for all real $\mathcal{L}$, provides natural multiplicative decomposition across channels, and time-normalizes across recall episodes of differing objective duration.

5 Memory Persistence Channel (M^*)

5.1 Fractional Relaxation Model

Standard exponential decay assumes smooth, monotone information loss over time. Empirical forgetting curves are better described by power functions (Wixted & Ebbesen, 1991; Wickelgren, 1974; Donkin & Nosofsky, 2012), reflecting multi-timescale consolidation processes (McClelland et al., 1995; Squire & Zola-Morgan, 1992). Trauma memory specifically exhibits long-tailed persistence inconsistent with exponential models (McNally, 2003; Brewin, 2011). We model autobiographical trace persistence via the Caputo fractional differential equation:

$$^{C}D_t^{\kappa} M(t) = -\mu M(t), \quad \kappa \in (0,1],\ \mu > 0,$$

where κ is the fractional order governing the sub-exponential decay exponent and μ is the decay rate.

5.2 Mittag-Leffler Solution

Theorem 5.1 (Memory Persistence Solution)**.** *The unique solution to the Caputo fractional relaxation equation with $M(0) = M_0$ is*

$$M(t) = M_0\, E_\kappa(-\mu t^\kappa),$$

where E_κ is the Mittag-Leffler function

$$E_\kappa(z) = \sum_{n=0}^{\infty} \frac{z^n}{\Gamma(\kappa n + 1)}.$$

Proof. Apply the Laplace transform using $\mathcal{L}\{^{C}D_t^{\kappa} f\}(s) = s^\kappa \tilde{f}(s) - s^{\kappa-1} f(0)$:

$$s^\kappa \tilde{M}(s) - s^{\kappa-1} M_0 = -\mu \tilde{M}(s) \implies \tilde{M}(s) = \frac{s^{\kappa-1} M_0}{s^\kappa + \mu}.$$

By the standard Laplace inverse identity (Mainardi, 2010):

$$\mathcal{L}^{-1}\left\{\frac{s^{\kappa-1}}{s^{\kappa}+\mu}\right\}(t) = E_{\kappa}(-\mu t^{\kappa}),$$

yielding $M(t) = M_0\, E_{\kappa}(-\mu t^{\kappa})$. □

5.3 Normalized Memory Density

Definition 5.2 (Normalized Memory Density)**.**

$$M^* = \frac{1}{t}\int_0^t M(s)\, ds.$$

Proposition 5.3 (Closed Form of M^*)**.**

$$M^* = M_0\, E_{\kappa,2}(-\mu t^{\kappa}),$$

where $E_{\kappa,2}(z) = \sum_{n=0}^{\infty} z^n/\Gamma(\kappa n+2)$ is the two-parameter Mittag-Leffler function.

Proof.

$$M^* = \frac{M_0}{t}\int_0^t E_{\kappa}(-\mu s^{\kappa})\, ds = \frac{M_0}{t}\sum_{n=0}^{\infty}\frac{(-\mu)^n}{\Gamma(\kappa n+1)}\int_0^t s^{\kappa n}\, ds = M_0\sum_{n=0}^{\infty}\frac{(-\mu t^{\kappa})^n}{\Gamma(\kappa n+2)} = M_0\, E_{\kappa,2}(-\mu t^{\kappa}).$$

The interchange of sum and integral is justified by uniform convergence of the Mittag-Leffler series on $[0, t]$ (Haubold et al., 2011). □

The memory information density contribution is $I_M = \beta M^*$. The Mittag-Leffler function interpolates between exponential decay ($\kappa = 1$) and pure power-law decay ($\kappa \to 0$), matching the empirical range of forgetting curves (Wixted & Ebbesen, 1991). For $\kappa < 1$, $E_{\kappa}(-\mu t^{\kappa})$ decays more slowly than any exponential, modeling the long-tail persistence observed in trauma memory (Brewin, 2011; McNally, 2003).

5.4 Parameter Locking

The fractional order κ is estimated via Mittag-Leffler kernel regression on cohort recall datasets. The decay rate μ is fixed from the same cohort. No adjustment of κ or μ is permitted after cohort-level estimation.

6 Affective Channel (E^*)

6.1 Self-Assessment Manifold

Affective state is captured via the Self-Assessment Manikin (SAM) (Bradley & Lang, 1994), yielding arousal $A \in [0, 1]$ and valence $V \in [-1, 1]$.

Definition 6.1 (Affective Charge)**.**

$$E^* = A \cdot \text{sign}(V) \cdot |V|^\lambda, \quad \lambda = 0.7 \textit{ (fixed)}.$$

The power law with $\lambda < 1$ implements compressive nonlinearity in valence magnitude, consistent with psychophysical scaling of emotional intensity (Russell, 1980; Bradley & Lang, 2000). The value $\lambda = 0.7$ is derived from cross-cohort SAM data. Re-estimation is permitted only when $n \geq 200$ within a new cohort.

By the SAM instrument bounds, $E^* \in [-1, 1]$ strictly. The affective information density contribution is:

$$I_E = \log(1 + E^*).$$

The logarithm implements diminishing marginal returns on affect, consistent with saturation in amygdala-driven arousal responses (LeDoux, 1996; Cahill & McGaugh, 1995).

The $(1+E^*)$ term and the factor of 2. In the assembled equation, the affective channel appears as $(1 + E^*)$. At maximum affect ($E^* = 1$), this term equals 2. At minimum affect ($E^* = 0$), it equals 1. This bounded multiplicative range is a direct consequence of the SAM instrument bounds—it is not a free parameter or an independently derived mechanistic quantity. The saturation prediction (P1) tests whether perceived duration follows a bounded function of arousal. It tests the qualitative form of saturation, not the precise value of 2, which is determined by instrument construction rather than neural theory.

7 Recursive Neural Entropy Channel (R)

7.1 Entropy Rate

Let $X_1, X_2, \ldots$ denote the neural state sequence (discretized EEG microstate or fMRI BOLD) during recall. Define block entropy $H_n = H(X_1, \ldots, X_n)$.

Definition 7.1 (Entropy Rate)**.**

$$h_R = \lim_{n\to\infty} \frac{H_n}{n}.$$

Definition 7.2 (Recursive Stability Coefficient)**.**

$$R = e^{h_R}.$$

Operationalization. Neural entropy rate is concretely measurable using EEG microstate entropy (Tononi et al., 1994; Tschacher et al., 2014) or multiscale entropy of BOLD signals (Costa et al., 2002). Both approaches have been applied in clinical PTSD populations (Lanius et al., 2010). The entropy rate estimator requires stationary, ergodic signals; minimum epoch length for reliable convergence is a measurement constraint acknowledged as an open specification task per neural phenotype.

The recursive information penalty is $I_R = \gamma \log R = \gamma h_R$. PTL models R as a functional driver of temporal compression: high R (complex, non-recursive dynamics) compresses T'; low R (tight recursive loops, as in rumination) dilates T'. Whether this functional relationship is causal or correlational cannot be established by the framework alone—causal determination requires experimental manipulation of R independent of other channels, which is not yet demonstrated. PTL makes the functional claim; the causal claim is the object of future experimental work.

The exponent γ is locked per neural phenotype from EEG subtype or fMRI connectivity cluster.

8 Narrative Coherence Channel (C)

8.1 Sequential Semantic Entropy

Recall transcripts are processed using sentence-level semantic embeddings rather than raw token distributions. This operationalization measures narrative discontinuity—the actual construct of interest—rather than vocabulary diversity, which is an artifact of token-level entropy.

Pipeline:

1. **Sentence embedding.** Each sentence of the recall transcript is encoded using a sentence transformer (e.g., SBERT; Reimers & Gurevych 2019), mapping each sentence to a vector in semantic space.

2. **Sequential KL divergence.** A Gaussian is fit to the embedding distribution of each sliding window of $k = 3$ consecutive sentences. KL divergence between consecutive windows is computed. High KL divergence indicates high narrative discontinuity (low coherence).

3. **Coherence score.** H_C is the mean sequential KL divergence across the transcript. Low H_C = coherent, linear narrative. High H_C = fragmented, disorganized recall.

Definition 8.1 (Narrative Coherence Coefficient)**.**

$$C = e^{H_C}, \quad H_C = \overline{D_{\mathrm{KL}}}(\textit{sequential semantic windows}).$$

The coherence penalty is $I_C = \log C = H_C$.

Validation requirement. The sequential KL divergence operationalization must be validated against established narrative coherence rating instruments (Reese et al., 2011) in a separate preregistered cohort before use in PTL prediction tests. This validation study precedes P3 and P5 experiments.

High H_C (fragmented narrative) compresses T' via the penalty. Low H_C (coherent narrative) dilates T'. This is consistent with event segmentation theory (Zacks et al., 2007; Radvansky & Zacks, 2017): richly segmented, coherent narratives generate more contextual change signals per unit time, expanding retrospective duration estimates (Jeunehomme & D'Argembeau, 2019; Block, 1990).

9 Assembly of the Canonical PTL Equation

9.1 Full-Form Equation

Substituting all channel definitions into the exponential action and retaining the full affective term:

$$\boxed{T' = \alpha \ \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma} \cdot C^{-1}.} \tag{1}$$

This is the canonical PTL equation. No approximation is applied. The full form holds across the complete affect range: in neutral recall ($E^* \approx 0$), the affective term contributes a factor of 1 and duration is governed by memory density and narrative structure; at maximum affect ($E^* = 1$), the affective term contributes a factor of 2, a bounded ceiling determined by SAM instrument construction.

No high-affect approximation. Earlier formulations of PTL applied the approximation $1 + E^* \approx E^*$, valid only when $E^* \gg 1$. Since the SAM bounds constrain $E^* \leq 1$, this approximation is never satisfied within the model's own parameter space. The approximation is retired. Equation (1) is the unique correct form.

9.2 Bounded Saturation

Empirically, perceived duration is bounded (Fraisse, 1984; Grondin, 2010). The $\tanh(\beta M^*)$ term imposes this ceiling: as $M^* \to \infty$, $\tanh(\beta M^*) \to 1$, bounding the memory contribution. Combined with the bounded affective factor $(1 + E^*) \in [1, 2]$ and positive R, C, the equation produces a finite T' for all parameter values.

9.3 Partial Derivatives

All five predictions follow from the signs of the partial derivatives of Equation (1):

$$\frac{\partial T'}{\partial M^*} = \alpha\beta \operatorname{sech}^2(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma} C^{-1} > 0 \tag{2}$$

$$\frac{\partial T'}{\partial E^*} = \alpha \tanh(\beta M^*) \cdot R^{-\gamma} C^{-1} > 0 \tag{3}$$

$$\frac{\partial T'}{\partial C} = -\alpha \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma} C^{-2} < 0 \tag{4}$$

$$\frac{\partial T'}{\partial R} = -\gamma\alpha \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma-1} C^{-1} < 0 \tag{5}$$

The complete parameter reference is given in Table 1.

Table 1: PTL Equation Parameters and Empirical Anchors

Symbol	Description	Empirical Anchor
M^*	Normalized memory density	Mittag-Leffler kernel regression; κ, μ fixed from cohort
E^*	Affective charge	SAM arousal/valence; $\lambda = 0.7$ fixed ($n \geq 200$ to revise)
R	Recursive stability	EEG microstate or multiscale entropy; γ locked per phenoty
C	Narrative coherence	Sequential semantic KL divergence; validated vs. coherence
α	Global scaling	Fixed at cohort level; no individual-level optimization
β	Memory sensitivity	Fixed at cohort level; no individual-level optimization
γ	Recursion exponent	Locked per neural phenotype
λ	Affective power exponent	Fixed at 0.7; re-derivable if $n \geq 200$ with Miller–Madow cor

10 Estimator Consistency

Proposition 10.1 (Entropy Rate Estimator Consistency)**.** *Let $\hat{h}_R = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}(x_i)$, where $\hat{p}$ is estimated via a consistent density estimator. Then $\hat{h}_R \to h_R$ almost surely as $N \to \infty$.*

Proof. By the law of large numbers, $\hat{h}_R \to \mathbb{E}[-\log p(X)] = h_R$ under stationarity and ergodicity of $\{X_i\}$ and consistency of $\hat{p}$ (Beirlant et al., 1997). □

Corollary 10.2. *$\hat{R} = e^{\hat{h}_R} \to R = e^{h_R}$ almost surely, by continuity of the exponential.*

Proposition 10.3 (Semantic Entropy Estimator Consistency)**.** *Let $\tilde{H}_C = \hat{H}_C + \frac{|V|-1}{2N}$ be the Miller–Madow corrected estimate. Then $\tilde{H}_C \to H_C$ in probability as $N \to \infty$.*

Proof. The Miller–Madow correction removes the leading $O(1/N)$ bias of the plug-in entropy estimator (Miller, 1955). Consistency follows from standard plug-in entropy estimation results (Paninski, 2003). □

Corollary 10.4. *$\hat{C} = e^{\tilde{H}_C} \to C$ in probability.*

Theorem 10.5 (Canonical PTL Estimator Consistency)**.** *The estimator*

$$\hat{T}' = \alpha \tanh(\beta M^*) \cdot (1 + E^*) \cdot \hat{R}^{-\gamma} \cdot \hat{C}^{-1}$$

satisfies $\hat{T}' \to T'$ in probability as sample sizes grow, provided $R > 0$ and $C > 0$.

Proof. $\hat{R}^\gamma \to R^\gamma$ by the continuous mapping theorem. $\hat{C} \to C$ similarly. Since $R > 0$ and $C > 0$, $\hat{R}^{-\gamma}\hat{C}^{-1} \to R^{-\gamma}C^{-1}$. The terms $\alpha \tanh(\beta M^*)$ and $(1 + E^*)$ are computed from cohort-fixed or directly measured quantities. Therefore $\hat{T}' \to T'$. □

11 Five Falsifiable Predictions

11.1 P1 — Saturation Invariance

Claim. There exists a finite biological ceiling on T'. No level of affective arousal or memory density produces unbounded subjective time dilation during autobiographical recall.

Mathematical basis. The tanh regularization imposes:

$$\lim_{M^* \to \infty} T' = \alpha \cdot 1 \cdot (1 + E^*) \cdot R^{-\gamma} C^{-1} < \infty$$

for any finite E^*, R, C. The affective ceiling $(1+E^*) \leq 2$ is set by SAM instrument bounds. P1 tests whether the *qualitative* saturating form is correct—whether perceived duration follows a bounded function of arousal rather than a monotone-increasing one. It does not test the precise numeric value of 2, which is determined by instrument construction.

Hypothesis 1 (Saturation Invariance). Retrospective duration estimates for high-arousal autobiographical events follow a saturating function of arousal intensity, not a monotone-increasing function. As E^* increases past a cohort-estimated threshold, additional increases produce diminishing marginal increases in $\hat{T}'$.

Operationalization. Administer standardized high-arousal autobiographical recall protocols to $n \geq 100$ participants across three arousal bands (low, moderate, high,

measured by SAM). Compute $\hat{T}'$ per band. Fit saturating (tanh-form) versus linear models. Predict: AIC favors the saturating model, with a statistically identifiable inflection point.

11.2 P2 — Trauma Persistence Asymmetry

Claim. Emotionally recursive memories decay according to Mittag-Leffler dynamics, producing long-tailed persistence. Old, emotionally charged memories retain more influence on M^* than recent neutral events, beyond what exponential decay predicts.

Mathematical basis. For $\kappa < 1$:

$$E_\kappa(-\mu t^\kappa) \sim \frac{1}{\mu\,\Gamma(1-\kappa)}\, t^{-\kappa} \quad (t \to \infty),$$

yielding power-law decay, slower than any exponential. This inversion—an older traumatic memory contributing more to current M^* than a recent neutral memory—is impossible under exponential decay.

Hypothesis 2 (Trauma Persistence Asymmetry). For matched initial encoding strength, $\hat{T}'$ for a traumatic event at lag τ_1 exceeds that for a neutral event at lag $\tau_2 < \tau_1$, and the difference increases as τ_1 grows. Exponential decay models predict the opposite ordering.

Operationalization. Recruit trauma-exposed and control participants. Administer recall probes for target traumatic and neutral events at various lags. Fit Mittag-Leffler and exponential decay models to $\hat{T}'$-versus-lag. Predict: Mittag-Leffler fit is superior (lower AIC) for trauma memories; exponential fit is superior or equivalent for neutral memories.

11.3 P3 — Narrative Causality

Claim. Modifying narrative structure C alters retrospective temporal estimates of the same autobiographical event. Narrative reorganization is a functionally independent variable in subjective time perception.

Mathematical basis. From Equation (4):

$$\frac{\partial T'}{\partial C} = -\alpha\, \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma} C^{-2} < 0.$$

T' is strictly decreasing in C. Reducing C (increasing narrative coherence, lower H_C) strictly increases T' for fixed M^*, E^*, R.

Hypothesis 3 (Narrative Causality). Experimental manipulation of narrative coher-

ence produces a statistically significant change in $\hat{T}'$ for the same autobiographical event, with direction consistent with $\partial T'/\partial C < 0$.

Operationalization. Within-subjects design: participants recall a moderately arousing autobiographical event twice, with an intervening narrative reorganization task (structured journaling increasing coherence). Measure $\hat{H}_C$ from transcripts pre- and post-intervention. Measure $\hat{T}'$ at both timepoints, controlling for M^* and E^*. Predict: $\Delta\hat{H}_C < 0$ produces $\Delta\hat{T}' > 0$, detectable at $n \geq 60$ per group.

11.4 P4 — Dissociation Paradox

Claim. High recursive entropy ($R \uparrow$) produces subjective time compression, not expansion. Dissociative states generate temporal contraction via entropic fragmentation of the recursive neural loop, directly contradicting arousal-only models that predict dilation.

Mathematical basis. From Equation (5):

$$\frac{\partial T'}{\partial R} = -\gamma\alpha \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma-1} C^{-1} < 0.$$

This produces a bifurcation:

- **Tight recursion regime** (R small): coherent rumination or intrusive replay. Low neural entropy. Loop closes. T' is large (time dilates).
- **Frayed recursion regime** (R large): dissociative processing. High neural entropy. Loop does not close. T' is small (time compresses).

Hypothesis 4 (Dissociation Paradox). Participants exhibiting dissociative symptoms (DES; Bernstein & Putnam 1986) produce higher $\hat{R}$ from EEG entropy measures during recall and report lower $\hat{T}'$ for matched arousal events compared to non-dissociative controls. PTSD re-experiencing (tight loop, low R) produces higher $\hat{T}'$ than dissociative PTSD (frayed loop, high R) at matched arousal.

Operationalization. Recruit three groups: healthy controls, PTSD with re-experiencing symptoms (PCL-5), and PTSD with dissociative symptoms (DES-Taxon; Waller et al. 1996). Administer standardized trauma-relevant recall. Record EEG; compute $\hat{R}$ from entropy rate. Match SAM ratings (E^*) across groups at $p > 0.20$. Predict: $\hat{R}_{\text{dissociative}} > \hat{R}_{\text{re-experiencing}}$ and $\hat{T}'_{\text{dissociative}} < \hat{T}'_{\text{re-experiencing}}$, controlling for E^*, M^*, and C.

11.5 P5 — Coherence–Dilation Trade

Claim. Two individuals with identical M^* and E^* experience different T' based solely on narrative coherence C. Narrative structure is a causally independent functional variable

in subjective time.

Mathematical basis. Fix $M^* = m$, $E^* = e$, $R = r$. Then:

$$T_1' - T_2' = \alpha \tanh(\beta m) \cdot (1 + e) \cdot r^{-\gamma} \cdot \left(\frac{1}{C_1} - \frac{1}{C_2} \right).$$

If $C_1 < C_2$ (Person 1 has higher narrative coherence), then $T_1' > T_2'$.

Hypothesis 5 (Coherence–Dilation Trade). Under conditions matching $\hat{M}^*$ and $\hat{E}^*$ across participants, $\hat{H}_C$ will be a statistically significant independent predictor of $\hat{T}'$, accounting for variance beyond arousal and memory strength.

Operationalization. Participants ($n \geq 80$) recall a standardized, moderately arousing film clip. Collect recall transcripts; compute $\hat{H}_C$ via sequential semantic KL divergence. Collect SAM ratings (E^*). Administer recognition battery to estimate $\hat{M}^*$ (d' as proxy). Compute $\hat{T}'$ as retrospective duration estimate divided by objective clip duration. Run hierarchical regression: Step 1 includes E^* and $\hat{M}^*$; Step 2 adds $\hat{H}_C$. Predict: $\hat{H}_C$ accounts for significant incremental variance ($\Delta R^2 > 0.05$, $p < .05$).

12 Discussion

12.1 Position Within Existing Literature

PTL integrates and extends three research traditions. From the retrospective memory tradition (Ornstein, 1969; Block, 1990; Block & Gruber, 2014), PTL inherits the principle that recalled duration depends on memory density, not clock time. The Fisher information metric formalizes "contextual change" as a precise mathematical quantity within the statistical manifold of neural recall states. From the affective timing tradition (Droit-Volet & Meck, 2007; Angrilli et al., 1997; Tse et al., 2004), PTL inherits the empirical relationship between arousal and duration distortion, formalizing it through the E^* channel with locked affective geometry. From the neural complexity tradition (Tononi et al., 1994; Friston et al., 1994), PTL inherits the use of entropy measures to characterize neural processing, applying them directly to the recursive dynamics of self-referential recall.

The key departure from all three traditions is the closed equation with no individual-level free parameters. Existing models either describe qualitative relationships or fit parameters post hoc. PTL constructs a specific functional form under justified assumptions and locks all parameters at the cohort level, enabling individual predictions without additional fitting.

12.2 Epistemic Status: Framework, Not Yet Theory

PTL is described throughout as a quantitative framework. The designation of "theory" is earned only upon empirical validation of the five predictions above. This distinction is not rhetorical caution—it is an accurate description of the framework's current epistemic status. A framework that generates specific, falsifiable, quantitative predictions in a domain where competitors generate none is scientifically valuable independent of whether those predictions ultimately survive. What PTL claims now: the modeling assumptions are justified, the equation is internally consistent, the predictions are operationalized and preregisterable, and the dissociation bifurcation is a novel contribution derivable from no competing framework. What PTL does not yet claim: empirical confirmation, causal demonstration of the R channel, or independent derivation of the Fisher metric outside the scope of Definition 1.

12.3 The Dissociation Prediction as a Theoretical Contribution

The P4 prediction is the most theoretically distinctive. Standard arousal-based accounts predict that high emotional intensity produces time dilation in all conditions. PTL predicts that the structure of the recursive neural process, not arousal alone, determines the direction of distortion. This bifurcation between tight-loop dilation and frayed-loop compression is not derivable from pacemaker-accumulator models (Treisman, 1963; Meck, 1983) or scalar timing theory (Gibbon, 1977). It requires the R channel and the specific functional form $R^{-\gamma}$ in the denominator. The empirical implication is that dissociative PTSD and re-experiencing PTSD should show opposite temporal distortion signatures at matched arousal—a sharp, falsifiable prediction with no competitor theory offering the same account.

12.4 Narrative Coherence as a Functional Variable

The C channel formalizes a claim implicit in narrative therapy (White & Epston, 1990) and expressive writing research (Pennebaker, 1997): the organization of a story about the past is not merely a reflection of processing but a functional variable that modulates the psychological duration of recalled events. PTL makes this explicit and quantitative. The $\partial T'/\partial C < 0$ relationship means any intervention that reduces narrative entropy—journaling, therapy, structured recall—should measurably increase the retrospective felt duration of the target event. This constitutes a testable, mechanistic account of one pathway through which narrative-based interventions alter memory.

12.5 Limitations

The Fisher metric application is conditional on Definition 1. Neural dynamics are high-dimensional, nonstationary, and often nonparametric; the degree to which the parametric statistical manifold approximation holds is an empirical question that validation studies should address.

The entropy rate h_R requires stationary, ergodic neural signals. Short EEG epochs during brief recall tasks introduce approximation error. Minimum segment length for reliable entropy estimation should be established per neural phenotype before clinical application.

The fractional order κ captures population-level differences in memory persistence but is locked at the cohort level. Hierarchical estimation at the individual level, without relaxing the no-individual-fitting constraint, is a direction for future methodological work.

The causal status of the R channel—whether neural entropy drives or merely correlates with temporal compression—is not established by the framework. Establishing causality requires experimental manipulation of R independent of other channels, which is a design challenge for future work.

All predictions require empirical validation. PTL is a framework under active construction; the predictions are falsifiable hypotheses, not established findings.

13 Conclusion

The Principle of Temporal Lensing establishes a closed, estimator-consistent, parameter-locked quantitative framework for subjective time distortion during autobiographical recall. Perceived duration is modeled as an informational path length on the Fisher manifold of neural–cognitive state space—a construction uniquely justified under reparametrization invariance given the statistical manifold assumption of Definition 1—governed by four channels: fractional memory density, affective charge geometry, recursive neural entropy, and narrative coherence. The canonical equation is:

$$T' = \alpha \ \tanh(\beta M^*) \cdot (1 + E^*) \cdot R^{-\gamma} \cdot C^{-1}.$$

No individual-level free parameters exist. No high-affect approximation is applied. Mathematical proofs establish the Mittag-Leffler solution for memory persistence, the closed form of normalized memory density, and the convergence of all estimators to their population targets.

Five falsifiable predictions follow from the equation's structure: a biological saturation ceiling on time dilation; power-law trauma memory persistence; functional modulation of retrospective duration by narrative reorganization; temporal compression under disso-

ciative high-entropy recursion; and independent coherence-based dilation under matched arousal and memory strength. Each prediction is operationalized with a preregisterable protocol and distinguishable from existing model predictions.

PTL does not describe time perception qualitatively. It computes a number, states where each parameter comes from, specifies what data would falsify it, and is honest about the assumptions on which it rests. The framework earns the title of theory when the predictions survive experiment.

Data and Code Availability

Repository: `https://github.com/NohMadLLC/FREv5`
License: CC-BY-NC-SA 4.0 (non-commercial scientific use only)

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